



Scalar Quantity:

All those quantities which requires only magnitude for complete description are called **scalar quantities** e.g. time, density, temperature and length etc.

Vector Quantity:

All those quantities which requires magnitude as well as direction for their complete description are called **vector quantities** e.g. weight, force, momentum, displacement, velocity etc.

Geometric Interpretation of Vector:

Geometrically, a vector is represented by a directed line segment $\overrightarrow{AB}$ with A its initial point and B its terminal point.

Magnitude of a Vector:

The magnitude or length or norm of vector $\overrightarrow{AB}$ is its absolute value and is written as $|\overrightarrow{AB}|$.

Unit Vector:

A unit vector $\hat{v}$ (read as v hat) of a given vector v is a vector with magnitude one and direction same as vector v . Mathematically

$$\text{Unit Vector} = \frac{\text{Vector}}{\text{Magnitude of Vector}}$$

$$\text{i.e. } \hat{v} = \frac{v}{|v|}$$

Null Vector:

A vector whose terminal point coincides with its initial point is called **null** or **zero vector**.

Negative of a Vector:

Two vectors u and v are called negative of each other, if they have same magnitude but opposite direction.

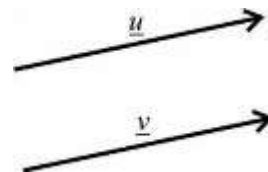
Multiplication of a Vector by a scalar (Number):

Multiplication of a vector v by a scalar ' n ' is a vector whose magnitude is n times that of ' v ' i.e. nv .

- (i) If n is positive, then v and nv are in the same direction.
- (ii) If n is negative, then v and nv are in opposite directions.

Equal Vectors:

Two vectors $\underline{u}$ and $\underline{v}$ are called equal vectors, if they have same magnitude and same direction.

**Parallel Vectors:**

Two vectors $\underline{u}$ and $\underline{v}$ are parallel if and only if they are non-zero scalar multiple of each other i.e.

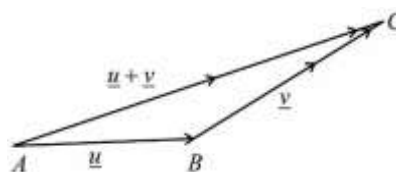
$$\underline{u} = \lambda \underline{v}, \lambda \neq 0$$

Triangle Law of Addition:

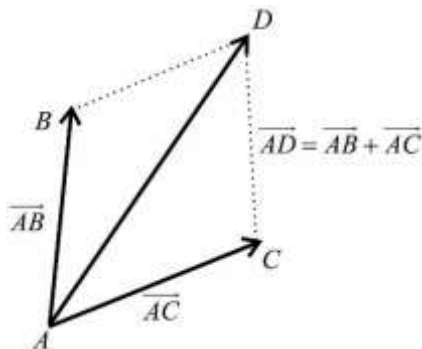
If two vectors $\underline{u}$ and $\underline{v}$ are represented by the two sides AB and BC of a triangle such that the terminal point of $\underline{u}$ coincide with the initial point of $\underline{v}$, then the third side AC of the triangle gives vector sum $\underline{u} + \underline{v}$, that is

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$$

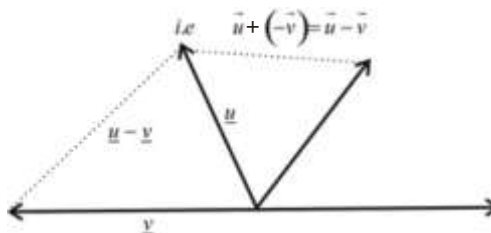
$$\Rightarrow \underline{u} + \underline{v} = \overrightarrow{AC}$$

**Parallelogram Law of Addition:**

If two vector $\underline{u}$ and $\underline{v}$ are represented by two adjacent sides $\overrightarrow{AB}$ and $\overrightarrow{AC}$ of a parallelogram as shown in the figure, then diagonal $\overrightarrow{AD}$ give the sum or resultant of $\overrightarrow{AB}$ and $\overrightarrow{AC}$, that is $\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{AC} = \underline{u} + \underline{v}$

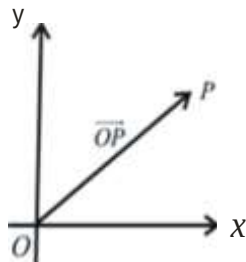
**Subtraction of two vectors:**

Let $\underline{u}$ and $\underline{v}$ are non-zero vector then subtraction of $\underline{v}$ and $\underline{u}$ is defined as the addition of $\underline{u}$ and $-\underline{v}$ i.e. $\underline{u} + (-\underline{v}) = \underline{u} - \underline{v}$

**Position Vector:**

A vector which describes the location of a point w.r.t origin is called position vector.

The vector, whose initial point is the origin O and whose terminal point P is called the position vector of the point P and is written as $\overrightarrow{OP}$



Vector in a Plane:

Let R be a set of real numbers. The Cartesian plane is defined to be the

$$R^2 = \{(x, y) : x, y \in R\}.$$

The Unit Vectors $\underline{i}$, $\underline{j}$:

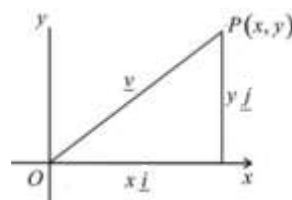
$\underline{i}$ and $\underline{j}$ are called unit vectors along x-axis and y-axis respectively.

They are written as

$$\underline{i} = [1, 0], \underline{j} = [0, 1]$$

$$|\underline{i}| = \sqrt{(1)^2 + (0)^2} = 1$$

$$|\underline{j}| = \sqrt{(0)^2 + (1)^2} = 1$$



A vector $\vec{v}$ can be written as

$$\underline{v} = [x, y] = [x, 0] + [0, y] = x[1, 0] + y[0, 1] = x\underline{i} + y\underline{j}$$

Similarly, sum of two vectors $\underline{u}$ and $\underline{v}$ can be written as

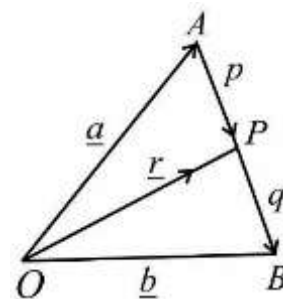
$$\underline{u} = [x_1, y_1], \underline{v} = [x_2, y_2]$$

$$\underline{u} + \underline{v} = [x_1 + x_2, y_1 + y_2] = (x_1 + x_2)\underline{i} + (y_1 + y_2)\underline{j}$$

The Ratio Formula:

Let A and B be two points whose position vectors are $\underline{a}$ and $\underline{b}$ respectively. If a point P divides AB in the ratio $p : q$, then the position vector of P is given by

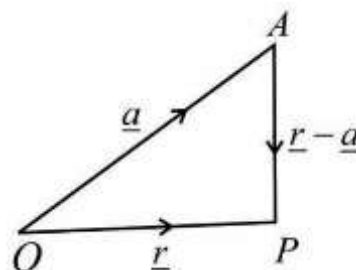
$$\underline{r} = \frac{q\underline{a} + p\underline{b}}{q + p}$$



Proof:

Given $\underline{a}$ and $\underline{b}$ are position vectors of the points A and B respectively. Let $\underline{r}$ be the position vector of the point P which divides the line segment $\overline{AB}$ in the ratio $p : q$. That is

$$m\overline{AP} : m\overline{PB} = p : q$$



so

$$\frac{m\overline{AP}}{m\overline{PB}} = \frac{p}{q}$$

$$\Rightarrow q(m\overline{AP}) = p(m\overline{PB})$$

Thus $q(\overline{AP}) = p(\overline{PB})$

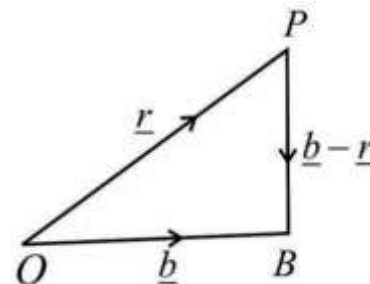
$$\Rightarrow q(\underline{r} - \underline{a}) = p(\underline{b} - \underline{r})$$

$$\Rightarrow q\underline{r} - q\underline{a} = p\underline{b} - p\underline{r}$$

$$\Rightarrow p\underline{r} + q\underline{r} = p\underline{b} + q\underline{a}$$

$$\Rightarrow \underline{r}(p+q) = q\underline{a} + p\underline{b}$$

$$\Rightarrow \underline{r} = \frac{q\underline{a} + p\underline{b}}{q+p}$$



Corollary:

If P is the midpoint of AB , then $p:q = 1:1$

$$\therefore \text{Position vector of } P = \underline{r} = \frac{\underline{a} + \underline{b}}{2}$$

Vectors in Space:

The set $R^3 = \{(x, y, z) : x, y, z \in R\}$ is called the 3-dimensional space.

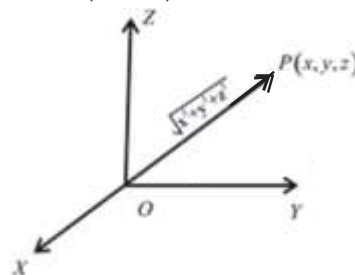
(i) Position Vector

The position vector of a point $P(x, y, z)$ in space, from the origin $O(0, 0, 0)$ is

$$\overline{OP} = x\underline{i} + y\underline{j} + z\underline{k}$$

The magnitude of $\overline{OP}$ is the distance of point P from the origin, i.e.

$$|\overline{OP}| = \sqrt{x^2 + y^2 + z^2}$$



(ii) The unit vectors $\underline{i}, \underline{j}, \underline{k}$

$\underline{i}, \underline{j}$ and $\underline{k}$ are called unit vectors along X, Y, Z axes respectively. They are written as:

$$\underline{i} = [1, 0, 0] \quad \underline{j} = [0, 1, 0] \quad \underline{k} = [0, 0, 1]$$

$$|\underline{i}| = \sqrt{(1)^2 + (0)^2 + (0)^2} = 1$$

$$|\underline{j}| = \sqrt{(0)^2 + (1)^2 + (0)^2} = 1$$

$$|\underline{k}| = \sqrt{(0)^2 + (0)^2 + (1)^2} = 1$$

A vector $\underline{v}$ can be written as:

$$\begin{aligned}\underline{v} &= [x, y, z] = [x, 0, 0] + [0, y, 0] + [0, 0, z] \\ &= x[1, 0, 0] + y[0, 1, 0] + z[0, 0, 1] \\ &= x\underline{i} + y\underline{j} + z\underline{k}\end{aligned}$$

Similarly, sum of two vectors $\underline{u}$ and $\underline{v}$ can be written as:

$$\begin{aligned}\underline{u} &= [x_1, y_1, z_1], \quad \underline{v} = [x_2, y_2, z_2] \\ \underline{u} + \underline{v} &= [x_1 + x_2, y_1 + y_2, z_1 + z_2] \\ &= (x_1 + x_2)\underline{i} + (y_1 + y_2)\underline{j} + (z_1 + z_2)\underline{k}\end{aligned}$$

(iii) Distance between two points in space:

The distance between two points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ in space is given by:

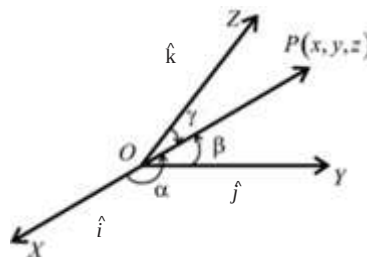
$$|\overline{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

(iv) Direction Angles and Direction Cosines of a vector:

Let $\underline{r} = \overline{OP} = x\underline{i} + y\underline{j} + z\underline{k}$ be a non-zero vector, let α, β and γ denote the angles formed between $\underline{r}$ and the unit coordinate vector $\underline{i}, \underline{j}$ and $\underline{k}$ respectively, such that $0 \leq \alpha \leq \pi$, $0 \leq \beta \leq \pi$ and $0 \leq \gamma \leq \pi$.

(i) The angles α, β, γ are called the direction angles.

(ii) The numbers $\cos \alpha, \cos \beta$ and $\cos \gamma$ are called direction cosines.



Important Result:

Prove that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

Proof:

$$\text{Let } \underline{r} = [x, y, z] = x\underline{i} + y\underline{j} + z\underline{k}$$

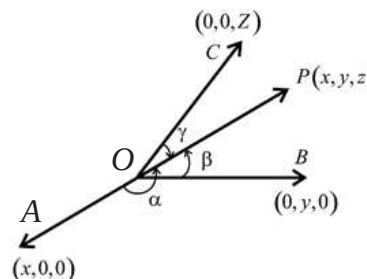
$$\therefore |\underline{r}| = \sqrt{x^2 + y^2 + z^2} = r$$

Then $\frac{\underline{r}}{|\underline{r}|} = \left[\frac{x}{r}, \frac{y}{r}, \frac{z}{r} \right]$ is the unit vector in the direction of the vector $\underline{r} = \overline{OP}$. It can be

visualized that the triangle OAP is a right triangle with $\angle A = 90^\circ$. Therefore in right triangle OAP ,

$$\cos \alpha = \frac{\overline{OA}}{\overline{OP}} = \frac{x}{r},$$

Similarly

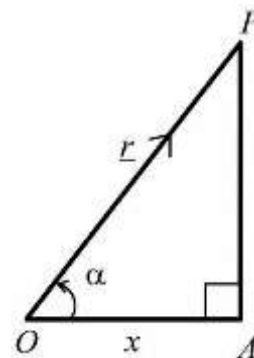


$$\cos \beta = \frac{y}{r}, \quad \cos \gamma = \frac{z}{r}$$

The numbers $\cos \alpha = \frac{x}{r}$, $\cos \beta = \frac{y}{r}$ and $\cos \gamma = \frac{z}{r}$ are called the

direction cosines of $\overrightarrow{OP}$.

$$\begin{aligned} \therefore \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma &= \frac{x^2}{r^2} + \frac{y^2}{r^2} + \frac{z^2}{r^2} \\ &= \frac{x^2 + y^2 + z^2}{r^2} = \frac{r^2}{r^2} = 1 \end{aligned}$$



The Scalar Product of two Vectors:

Definition: 1

The scalar or dot product of two vectors $\underline{u}$ and $\underline{v}$ in a plane or in space is

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta$$

where θ is the angle between $\underline{u}$ and $\underline{v}$ and $0 \leq \theta \leq \pi$.

The unit vectors $\underline{i}, \underline{j}, \underline{k}$:

$$(a) \quad \underline{i} \cdot \underline{i} = |\underline{i}| |\underline{i}| \cos 0^\circ = (1)(1)(1) = 1 \quad \text{as } \cos 0 = 1$$

$$\underline{j} \cdot \underline{j} = |\underline{j}| |\underline{j}| \cos 0^\circ = (1)(1)(1) = 1$$

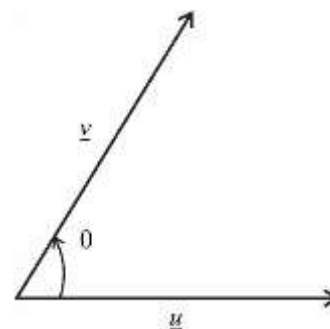
$$\underline{k} \cdot \underline{k} = |\underline{k}| |\underline{k}| \cos 0^\circ = (1)(1)(1) = 1$$

$$(b) \quad \underline{i} \cdot \underline{j} = |\underline{i}| |\underline{j}| \cos 90^\circ = (1)(1)(0) = 0 \quad \text{as } \cos 90^\circ = 0$$

$$\underline{j} \cdot \underline{k} = |\underline{j}| |\underline{k}| \cos 90^\circ = (1)(1)(0) = 0$$

$$\underline{k} \cdot \underline{i} = |\underline{k}| |\underline{i}| \cos 90^\circ = (1)(1)(0) = 0$$

$$(c) \quad \underline{u} \cdot \underline{v} = \underline{v} \cdot \underline{u} \quad (\text{Dot product of two vectors is commutative})$$



Definition: 2

(a) If $\underline{u} = a_1 \underline{i} + b_1 \underline{j}$ and $\underline{v} = a_2 \underline{i} + b_2 \underline{j}$ are two vectors in a plane, then the dot product of $\underline{u}$ and $\underline{v}$ is

$$\begin{aligned} \underline{u} \cdot \underline{v} &= (a_1 \underline{i} + b_1 \underline{j}) \cdot (a_2 \underline{i} + b_2 \underline{j}) \\ &= a_1 a_2 (\underline{i} \cdot \underline{i}) + a_1 b_2 (\underline{i} \cdot \underline{j}) + a_2 b_1 (\underline{j} \cdot \underline{i}) + (b_1 b_2) (\underline{j} \cdot \underline{j}) \\ &= a_1 a_2 (1) + a_1 b_2 (0) + a_2 b_1 (0) + b_1 b_2 (1) \\ \underline{u} \cdot \underline{v} &= a_1 a_2 + b_1 b_2 \end{aligned}$$

(b) If $\underline{u} = a_1\underline{i} + b_1\underline{j} + c_1\underline{k}$ and $\underline{v} = a_2\underline{i} + b_2\underline{j} + c_2\underline{k}$ are two vectors in space, then the dot product of $\underline{u}$ and $\underline{v}$ is

$$\begin{aligned}\underline{u} \cdot \underline{v} &= (a_1\underline{i} + b_1\underline{j} + c_1\underline{k}) \cdot (a_2\underline{i} + b_2\underline{j} + c_2\underline{k}) \\ &= a_1a_2(\underline{i} \cdot \underline{i}) + a_1b_2(\underline{i} \cdot \underline{j}) + a_1c_2(\underline{i} \cdot \underline{k}) + a_2b_1(\underline{j} \cdot \underline{i}) \\ &\quad + b_1b_2(\underline{j} \cdot \underline{j}) + b_1c_2(\underline{j} \cdot \underline{k}) + a_2c_1(\underline{k} \cdot \underline{i}) + b_2c_1(\underline{k} \cdot \underline{j}) + c_1c_2(\underline{k} \cdot \underline{k}) \\ &= a_1a_2(1) + a_1b_2(0) + a_1c_2(0) + a_2b_1(0) + b_1b_2(1) \\ &= b_1c_2(0) + a_2c_1(0) + b_2c_1(0) + c_1c_2(1) \\ \underline{u} \cdot \underline{v} &= a_1a_2 + b_1b_2 + c_1c_2\end{aligned}$$

Perpendicular (Orthogonal) Vectors:

If two vectors $\underline{u}$ and $\underline{v}$ are perpendicular, then

$$\begin{aligned}\underline{u} \cdot \underline{v} &= |\underline{u}| |\underline{v}| \cos 90^\circ = |\underline{u}| |\underline{v}| (0) \\ \underline{u} \cdot \underline{v} &= 0\end{aligned}$$

Angle between two vectors:

The angles between two vectors $\underline{u}$ and $\underline{v}$ is

$$(a) \quad \underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta$$

$$\therefore \cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$(b) \text{ If } \underline{u} = a_1\underline{i} + b_1\underline{j} + c_1\underline{k}$$

$$\underline{v} = a_2\underline{i} + b_2\underline{j} + c_2\underline{k}$$

$$\underline{u} \cdot \underline{v} = a_1a_2 + b_1b_2 + c_1c_2$$

$$|\underline{u}| = \sqrt{a_1^2 + b_1^2 + c_1^2} \quad |\underline{v}| = \sqrt{a_2^2 + b_2^2 + c_2^2}$$

$$\therefore \cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$\therefore \cos \theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Projection of one vector upon another vector:

Let $\underline{u} = \overrightarrow{OA}$ and $\underline{v} = \overrightarrow{OB}$ and θ be the angle between them. Where $0 \leq \theta \leq \pi$.

Draw $\overline{BM} \perp \overline{OA}$. Then $\overline{OM}$ is called the projection of $\underline{v}$ along $\underline{u}$. In right triangle OMB

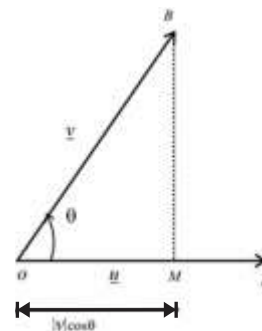
$$\cos \theta = \frac{\overline{OM}}{\overline{OB}}$$

$$\Rightarrow \overline{OM} = \overline{OB} \cos \theta$$

$$\therefore \overline{OM} = |\underline{v}| \cos \theta \quad (i)$$

$$\text{Also } \cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|} \quad (ii)$$

putting (ii) in (i)



$$\begin{aligned}
 \overline{OM} &= |\underline{v}| \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|} \\
 &= \frac{\underline{u} \cdot \underline{v}}{|\underline{u}|} \\
 &= \frac{|\underline{u}| |\underline{v}| \cos \theta}{|\underline{u}|} = |\underline{v}| \cos \theta
 \end{aligned}$$

i.e. projection of $\underline{v}$ along $\underline{u} = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}|} = |\underline{v}| \cos \theta$

Similarly.

Projection of $\underline{u}$ along $\underline{v} = \frac{\underline{u} \cdot \underline{v}}{|\underline{v}|} = |\underline{u}| \cos \theta$.

The Cross Product or Vectors Product of Two Vector:

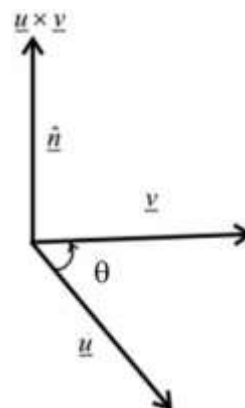
(i) Definition 1:

The vector or cross product of two vectors $\underline{u}$ and $\underline{v}$ in space is

$$\underline{u} \times \underline{v} = |\underline{u}| |\underline{v}| \sin \theta \underline{n}$$

where θ is the angle between $\underline{u}$ and $\underline{v}$ and $0 \leq \theta \leq \pi$. $\underline{n}$ is a unit vector perpendicular to the plane of $\underline{u}$ and $\underline{v}$ with direction given by the “right hand rule” stated below:

“If the fingers of right hand are curled in a direction from $\underline{u}$ towards $\underline{v}$, then the thumb will point in the direction of $\underline{n}$ which is $\underline{u} \times \underline{v}$. It is important to note that $\underline{u} \times \underline{v} \neq \underline{v} \times \underline{u}$, rather $\underline{u} \times \underline{v} = -(\underline{v} \times \underline{u})$.



(ii) The Unit Vectors $\underline{i}, \underline{j}, \underline{k}$:

$$(a) \quad \underline{i} \times \underline{i} = |\underline{i}| |\underline{i}| \sin 0^\circ \underline{n} = (1)(1)(0)\underline{n} = 0 \quad \because \sin 0^\circ = 0$$

$$\underline{j} \times \underline{j} = |\underline{j}| |\underline{j}| \sin 0^\circ \underline{n} = (1)(1)(0)\underline{n} = 0$$

$$\underline{k} \times \underline{k} = |\underline{k}| |\underline{k}| \sin 0^\circ \underline{n} = (1)(1)(0)\underline{n} = 0$$

$$(b) \quad \underline{i} \times \underline{j} = |\underline{i}| |\underline{j}| \sin 90^\circ \underline{k} = (1)(1)(1)\underline{k} = \underline{k} \quad \sin 90^\circ = 1$$

$$\underline{j} \times \underline{k} = |\underline{j}| |\underline{k}| \sin 90^\circ \underline{i} = (1)(1)(1)\underline{i} = \underline{i}$$

$$\underline{k} \times \underline{i} = |\underline{k}| |\underline{i}| \sin 90^\circ \underline{j} = (1)(1)(1)\underline{j} = \underline{j}$$

Definition 2:

If $\underline{u} = a_1 \underline{i} + b_1 \underline{j} + c_1 \underline{k}$ and $\underline{v} = a_2 \underline{i} + b_2 \underline{j} + c_2 \underline{k}$ are two vectors in space, then cross

$$\text{product of } \underline{u} \text{ and } \underline{v} \text{ is : } \underline{u} \times \underline{v} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$$

which is called the “Determinant formula” for $\underline{u} \times \underline{v}$.

Parallel Vectors:

If two vectors $\underline{u}$ and $\underline{v}$ are parallel, then

$$\underline{u} \times \underline{v} = |\underline{u}| |\underline{v}| \sin 0^\circ \underline{n} = |\underline{u}| |\underline{v}| (0) \underline{n}$$

$$\therefore \underline{u} \times \underline{v} = 0$$

Angle between two Vectors:

The angle θ between two vectors $\underline{u}$ and $\underline{v}$ is

$$\underline{u} \times \underline{v} = |\underline{u}| |\underline{v}| \sin \theta \underline{n}$$

$$\therefore \frac{\underline{u} \times \underline{v}}{\underline{n}} = |\underline{u}| |\underline{v}| \sin \theta$$

$$\therefore |\underline{u} \times \underline{v}| = |\underline{u}| |\underline{v}| \sin \theta$$

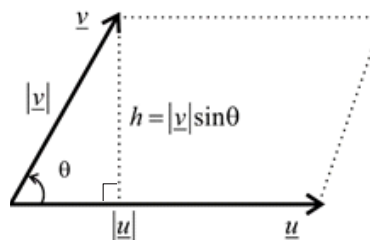
$$\therefore \sin \theta = \frac{|\underline{u} \times \underline{v}|}{|\underline{u}| |\underline{v}|}$$

Area of Parallelogram:

$$\text{Area of Parallelogram} = \text{Base} \times \text{Height}$$

$$= |\underline{u}| |\underline{v}| \sin \theta$$

$$\text{Area of Parallelogram} = |\underline{u} \times \underline{v}|$$

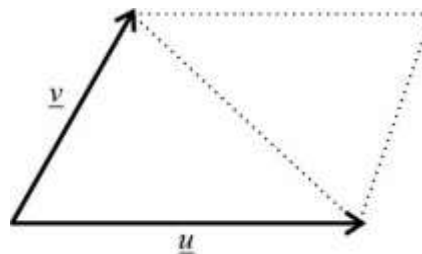
**Area of a Triangle:**

From the figure, it is clear Area of triangle

$$\text{Area of Triangle} = \frac{1}{2} (\text{Area of Parallelogram})$$

$$= \frac{1}{2} |\underline{u} \times \underline{v}| \text{ where } \underline{u} \text{ and } \underline{v} \text{ are}$$

vectors along to adjacent sides of triangle.

**Scalar Triple Product:**

For any three vectors $\underline{u}$, $\underline{v}$ and $\underline{w}$, the dot product of one vector with cross product of remaining two vectors is called “Scalar Triple Product” of vectors $\underline{u}$, $\underline{v}$ and $\underline{w}$. It is written as:

$$\underline{u} \cdot (\underline{v} \times \underline{w})$$

If $\underline{u} = a_1 \underline{i} + b_1 \underline{j} + c_1 \underline{k}$, $\underline{v} = a_2 \underline{i} + b_2 \underline{j} + c_2 \underline{k}$ and $\underline{w} = a_3 \underline{i} + b_3 \underline{j} + c_3 \underline{k}$ then

$$\underline{v} \times \underline{w} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$= \underline{i} (b_2 c_3 - c_2 b_3) - \underline{j} (a_2 c_3 - c_2 a_3) + \underline{k} (a_2 b_3 - b_2 a_3)$$

$$\begin{aligned}\underline{u} \cdot (\underline{v} \times \underline{w}) &= (a_1 \underline{i} + b_1 \underline{j} + c_1 \underline{k}) \cdot [(b_2 c_3 - c_2 b_3) \underline{i} - (a_2 c_3 - c_2 a_3) \underline{j} + (a_2 b_3 - b_2 a_3) \underline{k}] \\ &= a_1 (b_2 c_3 - c_2 b_3) - b_1 (a_2 c_3 - c_2 a_3) + c_1 (a_2 b_3 - b_2 a_3)\end{aligned}$$

$$\underline{u} \cdot (\underline{v} \times \underline{w}) = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

It is important to note that

$$\underline{u} \cdot (\underline{v} \times \underline{w}) = \underline{v} \cdot (\underline{w} \times \underline{u}) = \underline{w} \cdot (\underline{u} \times \underline{v})$$

The Volume of the Parallelepiped:

The Scalar triple product i.e. $\underline{u} \cdot (\underline{v} \times \underline{w})$ is **volume of a parallelepiped**. Hence it is a scalar.

The Volume of Tetrahedron:

$$\text{The volume of Tetrahedron} = \frac{1}{6} \underline{u} \cdot (\underline{v} \times \underline{w})$$

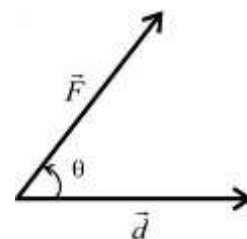
The properties of Scalar Triple Product:

- (i) If $\underline{u}$, $\underline{v}$ and $\underline{w}$ are coplanar then the volume of the parallelepiped is zero that is $(\underline{u} \times \underline{v}) \cdot \underline{w} = 0$
- (ii) If any two vector of scalar triple product are equal, then its values is zero i.e $[\underline{u} \ \underline{v} \ \underline{v}] = 0$

Work done by a Force:

If a constant Force $\vec{F}$ acts on a body, at an angle θ to the direction of motion, then work done by $\vec{F}$ is define to the product of the component of $\vec{F}$ in the , direction of the displacement and the distance that the body moves.

$$\text{Work done} = \underline{F} \cdot \underline{d} = (F \cos \theta) \underline{d} = \underline{Fd} \cos \theta$$



Moment of a Force (Torque):

The turning effect produced by a force is called “Torque” or “Moment” of that Force.

Moment = Perpendicular distance between point of application of force and point of rotation $\times$ Force applied.

$$\begin{aligned}\text{Moment of } \vec{F} \text{ about } O &= \overrightarrow{OP} \times \vec{F} \\ &= \vec{r} \times \vec{F}\end{aligned}$$

